Version-Bound Maintenance for Mutable Homomorphic Sparse Matrix–Vector Multiplication: A Fail-Closed Evaluation Boundary

> **Manuscript status:** Route C methods/boundary working draft. The primary one-shot qualification and a later, separately preregistered follow-up both selected their frozen stop routes. No comparative performance, complete-reference, security, or end-to-end claim is released. Evidence status last synchronized: 2026-08-31. A separately preregistered current-source deterministic conformance replication passed for one exact source and one fixed OpenFHE fixture; its scope is stated below.

Abstract

Compressed sparse layouts make homomorphic sparse matrix–vector multiplication practical for static patterns, but updates also change value placement, query- reorganization metadata, and encrypted-output-to-row mappings. Full recompression at each freshness boundary requires base-wide work; auxiliary components trade publication work for query work and observable structure.

We present an update-aware layer around the published Compressed Sparse Sorted Column (CSSC) representation. Freshness-bounded Publication Windows bind each query to an immutable matrix version, global-column metadata, and a private RowMap-sensitive reconstruction plan. For multi-component layouts, the plan distinguishes overlaps from disjoint blocks and drives overlap-scoped one-time zero-sum masking with persistent no-reuse bindings. A fixed-segment delta lets the Cloud execute a public page-wide schedule while the client privately merges segment leaders. A typed bundle binds the CSSC base, delta, operands, routes, and operation graph and fails closed on version or plan substitution.

Our frozen evaluation protocol was specified to compare three fixed maintenance mechanisms under the same versioned event streams, query schedules, public parameters, and initial states. An admitted execution would have separated state-transition costs, exact typed operation counts, type-derived serialized-byte bounds, a narrowly registered query-linearity projection, and native OpenFHE observations. The frozen matrix specified six synthetic shards, four ordered-event shards derived from one SNAP source object, and six native OpenFHE cases. Independent replay and terminal admission would have had to accept exactly the resulting 17 pre-aggregate artifacts before analysis could release any empirical sentence.

The primary preregistered, permanently non-admissible qualification did not reach its combined guard within the frozen computational deadline. A later, separately preregistered follow-up passed five exact-S2 authority-false controls, but its sole provider run was cancelled during hosted-runner setup after the external controller could not establish the watcher-admission binding. No repository checkout, registered-seed step, scientific stage, or artifact occurred. Under the frozen one-shot rules, both paths terminated without a formal-dispatch capability; the acquisition and 16-unit formal campaign were not started. Without reopening either performance lineage, a separately preregistered current-source functional replication subsequently passed its eight-case, 35-record deterministic contract and one fixed pinned- OpenFHE whole-query witness. The witness decrypted to the prespecified sparse vector and matched two independent plaintext oracles. We report the protocol, its definition-level functional obligations, the source/evidence separation, and these fail-closed evaluation boundaries, plus that version-bound single-fixture result. We do not convert partial qualification execution, runner setup, prerequisite controls, or the conformance witness into strategy-cost or native-OpenFHE performance evidence.

Keywords: homomorphic encryption; sparse matrix– vector multiplication; mutable sparse matrices; cryptographic engineering; reproducible evaluation

1. Introduction

Encrypted sparse linear algebra sits at an awkward boundary between cryptography and data layout. Homomorphic encryption hides values but makes data movement and permutation expensive. Sparse formats reduce arithmetic on zeros, yet they also turn a matrix’s support into layout metadata and require a client to align the encrypted query with physical value lanes. CSSC addresses the static case by sorting rows, packing capacity-fitting rectangles, and retaining the metadata needed to recover logical outputs. Its own stated limitation is a static sparse pattern (Gao et al. 2026).

A mutable matrix cannot be handled by changing values alone. An insertion may consume padding, create an overflow component, change a row permutation, or force new query-reorganization metadata. A deletion may leave reusable physical capacity without changing the logical matrix dimension. Meanwhile, a query must not combine a new matrix component with an old column-index array or reconstruct shares using a plan from another version. These are system-level consistency requirements, not details of the underlying homomorphic primitive.

The naive policy republishes a fully recompressed CSSC layout after every batch. Alternative policies retain slack, patch local regions, append a delta, or periodically fold auxiliary state into the base. Their relative merit depends on the update/query ratio, freshness deadline, layout history, communication bandwidth, and the exact encrypted output path. Evaluating each window from a fresh initial layout, selecting with held-out information, or omitting returned ciphertexts and masking work can confound a comparison.

This paper therefore asks a narrower, auditable question: how should mutable CSSC state be published, queried, reconstructed, and compared without breaking version semantics or overstating evidence?

Contributions

1. **Version-bound publication semantics.** We define Publication Windows and a commit rule that binds logical state, CSSC components, global-column query metadata, the reconstruction plan, and prepared queries to one version.
2. **Explicit multi-component reconstruction.** We introduce a private RowMap-sensitive OutputPlan that distinguishes overlap summation, disjoint block concatenation, and implicit zeros.
3. **Scoped masking integration.** We derive overlap-only F1-M operands from the plan and bind every random mask to a persistent, reserve-before-sample five-field identity. The underlying canceling-mask idea is prior art; the contribution claimed here is its versioned CSSC integration.
4. **A designated strong fixed-segment reference path.** We define public fixed-width segment/page shapes, cloud-side leader reduction, private leader merging, and a deterministic whole-query execution bundle. We do not claim a new HE primitive or a formal security proof.
5. **Causal, fail-closed evaluation.** We freeze three strategy identities rather than train an online selector, advance independent states through identical deterministic streams, separate measured and projected quantities, and bind every reportable claim to independently replayed, digest-addressed evidence.

2. Background and Related Work

2.1 Static encrypted sparse matrix–vector multiplication

CSSC is the direct substrate of this work and must be credited for row sorting, left-aligned sparse rectangles, Value, global ColumnIndex, RowMap, chunked query reorganization, and static aggregation (Gao et al. 2026). Our implementation is an independent pseudocode-derived reconstruction; we did not use or verify an author implementation. Static encrypted sparse linear algebra already spans Lodia's low-diagonal SpMV (Yu et al. 2025), CipherSkip's encrypted-value-and-index arbitrary-shape SpGEMM (Xiong et al. 2026), diagonal-packing reordering (Mutluergil et al. 2026), ciphertext–ciphertext SpMSpM on CPU and GPU (Ferguson et al. 2025; D'Agata et al. 2026), and the encrypted-index Scatter–Gather–Apply design reported by the public SparseE program abstract (Wei et al. 2026). Rhombus separately shows plaintext-matrix/encrypted-vector two-party MVM with additive output shares (He et al. 2024). These works rule out broad claims of first encrypted indices, first sparsity-aware FHE multiplication, first ciphertext–ciphertext sparse multiplication, or first use of random sharing to split an MVM output. Our claimed gap is the narrower mutable CSSC publication and reconstruction contract; it is not a new static packing or cryptographic primitive.

Our source audit of CSSC (Gao et al. 2026) finds a non-power-of-two ambiguity in the printed aggregation pseudocode. We use a corrected totalSum-compatible stored-power/prefix graph that preserves the paper-derived abstract count floor(log2 w) + popcount(w) - 1. We present this as a compatibility correction, not as a new aggregation algorithm or verified author-code behavior; the totalSum connection is grounded in the HElib SIMD reduction literature (Halevi and Shoup 2014).

The closest methods are not interchangeable experimental baselines.

Table 1. Closest method families and the boundary of this work.

Work	Principal representation or setting	Mutable support?	Relation to this paper
CSSC (Gao et al. 2026)	Sparse-coordinate compression with ColumnIndex/RowMap	No	Direct static substrate; its published layout and reorganization are not our contribution.
Lodia (Yu et al. 2025)	Low-diagonal decomposition for batched FHE SpMV	Not described	Establishes earlier sparsity-aware encrypted SpMV; different decomposition and leakage/cost interface.
CipherSkip (Xiong et al. 2026)	Encrypted values and indices for arbitrary-shape SpGEMM and chained products	No matrix-state publication protocol	Its server-side alignment concerns encrypted intermediate products, not insert/delete/update of a published sparse layout.
SparseE (Wei et al. 2026)	Encrypted-index Scatter–Gather–Apply with a permutation/expansion accelerator	Not established by the public descriptions	Establishes an encrypted-index hardware co-design boundary; as of 2026-08-30, an official program abstract and a BUAA institutional summary were public, but no public full text, DOI, or public software repository was located.
Diagonal Packing (Mutluergil et al. 2026)	Row/column reordering to reduce occupied cyclic diagonals	Not described	HE-aware static compiler optimization; no mutable CSSC publication or reconstruction is described.
Ferguson et al. (Ferguson et al. 2025)	CKKS ciphertext–ciphertext SpMSpM with public sparse metadata	Not described	Demonstrates prior FHE sparsity exploitation under a PPML workload; different operation, scheme, and small square matrices.
D'Agata et al. (D'Agata et al. 2026)	GPU/FIDESlib CKKS ciphertext –ciphertext SpMSpM with public sparse metadata	Not described	Extends the same static design space to GPU execution; a dynamic SpMV publication protocol is not described.
Rhombus (He et al. 2024)	Plaintext-matrix/encrypted-vector MVM with additive output shares	Not described	A different two-party PPML protocol; the random-share primitive is prior art, while versioned overlap binding is our narrower integration claim.
Damie et al. (Damie et al. 2025)	Secret-shared sparse matrix multiplication in MPC	Not described	Shows secure sparse arithmetic outside FHE; incomparable without a common threat and cost model.

Work	Principal representation or setting	Mutable support?	Relation to this paper
This work	Version-bound CSSC base plus designated maintenance components; formal evaluation stopped at qualification	Yes	Studies publication, query recompilation, private reconstruction, and causal-cost contracts; claims no static-format, primitive, or comparative-performance novelty.

2.2 Mutable sparse layouts

Reserved slack, padding reuse, COO/HYB overflow, row-local growth, base/delta organization, periodic compaction, and format selection all have substantial plaintext sparse-matrix and storage-system precedent, including SELL-C-sigma (Kreutzer et al. 2014), GPU HYB/COO practice (Bell and Garland 2008), LSM-trees (O'Neil et al. 1996), Dynamic-CSR (King et al. 2016), and dynamic format selection (Stylianou and Weiland 2022). They are comparison families in this paper, not independent novelty claims. Our narrower concern is their interaction with encrypted query preparation, version consistency, private reconstruction, and complete cost accounting.

2.3 Dynamic encrypted data systems and masking

Dynamic SSE, encrypted databases, ShieldDB, and oblivious transaction systems already study updates, client state, epochs, padding, and leakage (Cash et al. 2014; Vo et al. 2023; Crooks et al. 2018). They do not instantiate the same ciphertext–ciphertext CSSC query, but they prevent broad claims that batching or versioned updates are new. d-DSE directly shows that update-volume leakage remains a separate problem and that padding can impose large storage and communication costs (Liu et al. 2024). CKKS-Auth Tree uses versioned root commitments and timestamps to detect stale or replayed verification objects after encrypted-metadata updates (Chen et al. 2026). Neither work implements mutable SpMV, but together they rule out novelty claims for volume hiding, versioned commitments, freshness checks, or replay rejection in isolation. Fixed segments, dummy work, and visible schedules are therefore described through an explicit leakage surface, not as a generic update-leakage solution.

Canceling one-time masks also predate this work (Bonawitz et al. 2017). We therefore state the exact leakage and recipient roles and avoid the terms simulation-secure or prevents leakage without a separate proof. Authorized homomorphic database updates also predate this work and delimit any claim about encrypted mutability (Parbat and Chatterjee 2023).

3. System and Threat Model

The system has Client A, which owns the mutable matrix, updates, CSSC metadata, component ColumnIndex arrays and RowMaps, the complete OutputPlan, and the F1-M reservation/sampling ledger; Client B, which owns the query and secret key and receives the result; and a semi-honest Cloud evaluator. Client A and Client B are both authorized to read the complete RowMap-sensitive OutputPlan;

private means private from the Cloud, not private between the two clients. Client A sends the versioned column metadata and complete plan to Client B, samples and encrypts F1-M operands under Client B's public key, and never sends the individual mask plaintexts to Client B. At most one party is corrupted and the Cloud does not collude with either client. The model excludes malicious behavior, adaptive corruption, availability, side-channel, traffic-analysis, and collusion claims.

Client B is authorized to learn the versioned global column indices, component RowMaps, complete reconstruction plan, and final result. The Cloud is authorized to learn public parameters, ciphertext shapes and counts, public page/segment shapes, the operation schedule, opaque identifiers, query/version identifiers, and plan digests. The interface classifies matrix/query plaintexts, global column metadata, RowMaps, full plans, mask plaintexts, and unblinded component outputs as forbidden-to-Cloud fields, and the typed serializers do not emit them. This is an access-control and serialization requirement, not a proof that all leakage is prevented or that the implementation realizes a simulation-based notion.

4. Design

Figure 1 shows the protocol boundary. Client A owns matrix publication, the complete plan and private metadata, constructs the version commit, and prepares the encrypted F1-M operands. Client B owns the query and key set, uses the version-matched column metadata to gather and encrypt query values, and uses its copy of the complete plan to recover logical coordinates. The Cloud receives only the typed public program, encrypted operands, and permitted identifiers. The diagram is an interface summary, not a claim that the listed metadata is cryptographically hidden from every side channel.

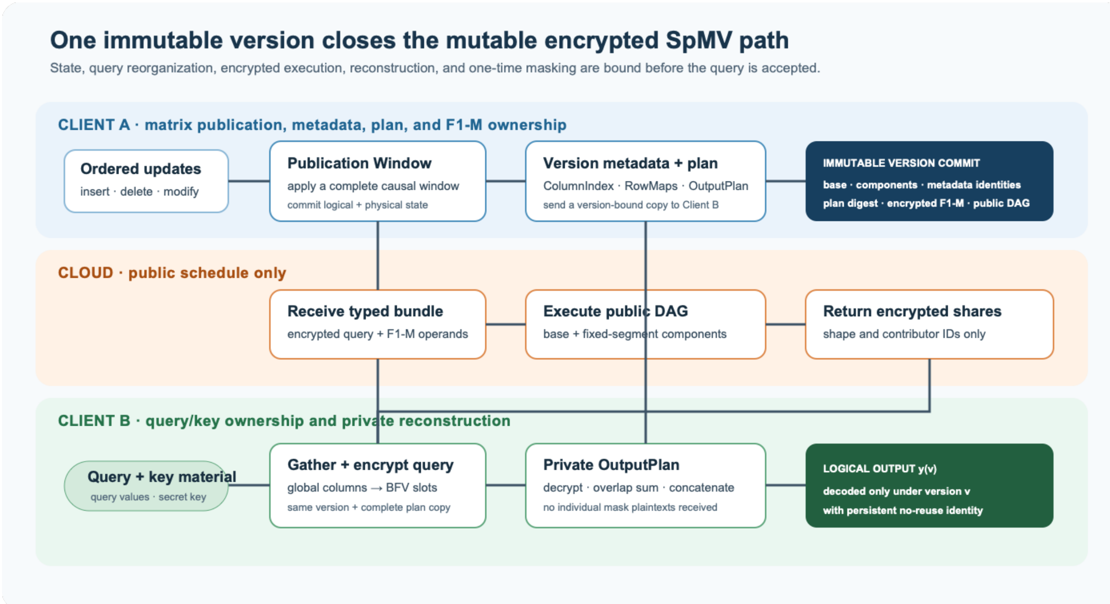

Figure 1. Version-bound mutable CSSC publication, execution, and private reconstruction.

4.1 Publication Windows

Updates accumulate until a query arrival, freshness deadline, microbatch bound, or explicit publication event closes the current window. The transition first applies every net update to a candidate logical state, validates matrix and value bounds, constructs or patches physical components, decodes them back to the same logical state, and only then publishes the next version. A failed transition cannot partially advance the visible state.

For the publication experiment, accepted raw events form atomic scheduling groups. All visible SET transitions are applied first, followed by a payload-free logical clock TICK and then the group's exactly scheduled queries. The TICK is emitted even for clipped no-ops, so freshness advances without fabricating an update. Before a group at logical time t is processed, any pending half-open window whose deadline is at or before t is closed; the deadline group therefore belongs to the next window. No close may split the group's SET→TICK→query sequence. One deterministic bound-one ternary query vector is frozen per paired trace unit and reused across candidates and cells. Its seed, generation method, and values are public, so it is a known-answer correctness/cost control rather than evidence for a production-query distribution, cryptographic randomness, or query-plaintext confidentiality.

4.2 Versioned query reorganization

Each CSSC value lane is paired with a parallel original matrix-column identifier or a padding sentinel. Client A sends versioned plaintext column metadata to Client B. Client B gathers one aligned vector per encrypted value chunk and encrypts it. The matrix column domain is independent of the ciphertext-slot domain; reducing a global column identifier modulo the slot count changes the function and is forbidden.

4.3 OutputPlan and F1-M

The plan maps (component, output block, physical lane) to a logical output coordinate. Client B initializes the public-length result to zero, reorders each decrypted share, sums only equal logical coordinates, and concatenates disjoint horizontal blocks. A coordinate with no physical contributor is an implicit zero.

For contributor multiplicity greater than one, Client A samples a modular zero-sum mask tuple. The binding (query, version, plan digest, component, output block) is reserved in the durable SQLite ledger before sampling. Duplicate use is rejected within that ledger model. Repository rollback, database cloning or compromise, and cross-device coordination are outside the present evidence. A disjoint strong-path return uses an encrypted-zero dummy to retain a uniform visible operation position; it is not a random mask and is accounted separately.

4.4 Strong fixed-segment delta

The strong path retains a real CSSC base and places overflow into fixed-width power-of-two segments, currently frozen to c=128. A segment contains entries from one logical row. The public typed plan uses uniform shapes and opaque ordinal IDs, while the Cloud-visible leakage also includes published counts, schedule and timing, query/version identifiers, and binding digests. We do not claim segment unlinkability. The Cloud multiplies aligned value/query ciphertexts, applies the fixed reduction schedule, selects segment leaders, adds one F1-M operand per return, and returns page shares. Client B uses the private plan to merge leaders that map to the same logical row.

The current policy does not authorize cloud-side merging of segments that share a logical row. Same-row-equivalence fields are carried only in Client B’s typed plan; this is an interface access-control requirement, not a proof that segment relationships cannot be inferred. The publication identity for folding/compaction is frozen separately; changing it creates a different candidate.

The segment width 128 is fixed before real-stream evaluation rather than tuned: it yields a seven-stage power-of-two reduction, places 32 complete segments in the 4,096-lane effective domain, and is the exact 127-active-plus-one-padding boundary covered by the pinned witness. We make no optimality or cross-segment-width claim.

4.5 Definition-level functional propositions

The following statements are conditional functional propositions about the frozen typed interfaces and algorithms. They are not empirical conclusions and not a malicious- or simulation-security theorem. The exact-S1 source-conformance record maps every premise below to the corresponding validator, state transition, reconstruction routine, or ledger transition; the registered tests exercise legal cases and substitutions but do not replace the proof arguments.

P1: binding soundness

For publication version v, let

$$P^{(v)} = \left(s^{(v)}, \{ C_k^{(v)} \}_{k=1}^{K_v}, \{ CI_k^{(v)} \}_{k=1}^{K_v}, \{ RM_k^{(v)} \}_{k=1}^{K_v}, O^{(v)}, Q^{(v)}, B^{(v)} \right),$$

where $s^{(v)}$ is the canonical logical-state identity binding the logical matrix $A^{(v)}$, C_k are ordered physical components, CI_k and RM_k are their global-column and row mappings, O is the private OutputPlan, Q is the prepared-query binding, and B is the immutable typed identity and digest bundle. Let $B^{(v,q)}$ be the authoritative binding for query q. The registered acceptance predicate has the shape

$$Accept(X; B^{(v,q)}) = \bigwedge_{f \in F} [X_f = B_f^{(v,q)}] \wedge \bigwedge_{o \in O} [SHA256(bytes_X(o)) = B_o^{(v,q)}],$$

where F enumerates every typed version, query, component, mapping, plan, parameter, and execution identity, and O enumerates every retained canonical byte object. Hence acceptance implies equality for every enumerated field and rehashed object. The capability and formal-artifact paths are downstream of this conjunction, so a failed conjunct cannot mint them.

The proof is a case analysis over the closed predicate: each field class occurs as a necessary conjunct, retained bytes are rehashed inside the private admission process, and no earlier branch reaches the issuer. This establishes a functional binding property for ordinary inputs and malformed/stale substitutions. It says nothing about deliberate SHA-256 collisions, compromised hosts, or side channels.

P2: multi-component reconstruction

Let $A_k^{(v)}$ be the physical-output-row by global-logical-column matrix represented by component k, and let E_k embed its declared physical output rows into the logical matrix domain. The admitted decomposition is complete when

$$A^{(v)} = \sum_{k=1}^{K_v} E_k \left(A_k^{(v)} \right) (mod\, t).$$

Assume each component execution decrypts to $A_k^{(v)} x$ in its declared lanes, every lane has exactly its declared logical coordinate, the OutputPlan is total, and P3 masks cancel. For logical row r, the reconstruction operator sums all contributors mapped to r, places disjoint blocks in logical order, and uses zero when the contributor set is empty. Therefore

$$R_{O^{(v)}} \left(\{ A_k^{(v)} x \}_{k=1}^{K_v} \right)_r = \sum_{k=1}^{K_v} \left(E_k (A_k^{(v)}) x \right)_r.$$

By completeness of the admitted decomposition,

$$\sum_{k=1}^{K_v} \left(E_k \left(A_k^{(v)} \right) x \right)_r = \left(A^{(v)} x \right)_r (\operatorname{mod} t).$$

The equality is coordinatewise: overlap is ordinary addition in $\mathbb{Z}_t$, disjoint blocks have disjoint images under the embeddings, and an unmaterialized row contributes the additive identity. It holds for any admitted number of components under the stated completeness and totality premises; the finite tests only check the implementation of those premises.

P3: mask cancellation and ledger-scoped no-reuse

For an overlap group $G_r = (k_1, \ldots, k_g)$ in canonical contributor order with $g \geq 2$, Client A samples

$$m_{r,k_1}, \ldots, m_{r,k_{g-1}} \xleftarrow{u.a.r.} \mathbb{Z}_t, m_{r,k_g} = -\sum_{i=1}^{g-1} m_{r,k_i} (\operatorname{mod} t).$$

Thus $\sum_{k \in G_r} m_{r,k} = 0 (\operatorname{mod} t)$, and P2 reconstruction removes the random operands without changing the logical output coordinate. Groups of size zero or one receive no random tuple; an encrypted-zero dummy is a separate accounted object.

For no-reuse, the durable state machine is

$$unseen \rightarrow reserved \rightarrow prepared \rightarrow consumed.$$

One transaction reserves every five-field key (query, version, plan digest, component, output block) before the first random sample. A prepared batch additionally binds the private plan, execution binding, modulus, operand commitments, and a unique token; verification consumes that token atomically. Uniqueness constraints plus exact terminal closure reject a duplicate key or token, commitment drift, orphan record, or second consumption. The evaluation-lane digest includes the unit-attempt ordinal, so the single provider-replacement allowance cannot reuse the same reservation identity. This is an invariant of one uncompromised durable SQLite ledger, not a rollback, cloning, compromise, or cross-device guarantee.

P4: fixed-segment reconstruction

For a logical row r, partition its auxiliary post-multiplication lane values $a_{r,c} x_c$ in canonical entry order into J_r segments $S_{r,1}, \ldots, S_{r,J_r}$ of $c = 128$ lanes, so every populated $S_{r,j}[\ell]$ is one such $a_{r,c} x_c$ value, and pad only the final unused suffix with zeros. The seven-stage public reduction places

$$L_{r,j} = \sum_{\ell=0}^{127} S_{r,j}[\ell]$$

in the predetermined leader lane. The private OutputPlan maps every leader back to r, so Client B obtains

$$\sum_{j=1}^{J_r} L_{r,j} = \sum_{j=1}^{J_r} \sum_{\ell=0}^{127} S_{r,j}[\ell],$$

which is exactly the auxiliary contribution to row r; adding the CSSC-base contribution and applying P2 gives the direct logical product. The proof is an induction on J_r: one segment is the fixed reduction identity, and appending a row-owned segment adds exactly its leader sum without changing prior leaders. The 127/128/129, tombstone, padding, disjoint, and overlap cases are boundary tests of this construction, not its general proof and not evidence that 128 is an optimal width.

5. Implementation and Evidence

At historical baseline commit fcb00e0d, the Python implementation supplied typed persistent states, a common query compiler, exact operation graphs, private operand/route bindings, a plaintext oracle, SQLite no-reuse commitments, canonical artifacts, and a separate deterministic replay validator. The encrypted path uses BFV-family batching, whose original scheme lineage is Fan–Vercauteren (Fan and Vercauteren 2012), through OpenFHE 1.5.1 pinned by source commit (OpenFHE Development Team 2026).

At that historical baseline, R0 passed 750 tests and the Phase 2 whole-query witness passed in run 32581653504. The witness exercises a 4096-by-8193 fixture, global-column anti-aliasing, the 127-of-128 segment boundary, the unused second BFV batching row, explicit no-relinearize/relinearize transitions, random and dummy F1-M operands, and equality with both typed and direct plaintext oracles. Its GitHub Actions artifact-wrapper SHA-256 digest is recorded in the version-bound claim-ledger supplement (c5f44b0c...4711afe).

The separately preregistered current-source replication retained that exact fixture, expected output, OpenFHE pin, and witness specification while binding them to reviewed source 844fb062d78f5095f14599c6c71a27cb6034f001 through lightweight tag current-source-e4-conformance-20260831-v1. Its only current- source run, 33386130654, completed successfully and produced artifact 9755741401. A clean controller independently verified the raw provider ZIP, the exact 19-file set, strict internal checksums, literal RFC 6901 pointer expectations, and every source digest in PROVENANCE.json against a fresh detached tag checkout. Section 7.6 states the bounded result and its exclusions.

This evidence is deliberately narrow. It does not establish complete candidate costs, candidate registration, mixed-circuit parameter safety, security, performance, or an end-to-end deployment.

Table 2. Definition-level obligations and their Route C disposition.

Obligation	Frozen implementation and review boundary	Claim permitted in this manuscript
P1: exact binding	Typed version/query/plan/payload identities, fresh byte rehashing, independent replay, and source-conformance mapping	Conditional binding proposition and fail-closed interface design; no end-to-end admission claim
P2: multi-component reconstruction	Private OutputPlan, direct plaintext oracle, overlap/concatenation/implicit-zero checks	Definition-level reconstruction proof under the admitted-plan premises; no formal-run result
P3: F1-M cancellation and no-reuse	Modular cancellation proof plus durable reserve-before-sample SQLite state machine	Cancellation and ledger-local invariant only; no simulation-security, rollback, cloning, or cross-device claim
P4: fixed 128-lane segment path	Seven-stage reduction proof and 127/128/129 boundary tests	Correctness proposition for the frozen segment construction; no optimal-width or performance claim
Evidence release	S1/S2 Behavior Sets, independent guards, terminal controller, and exact object-count contract	Engineering/source conformance and the Route C stop decision only; no strategy-cost evidence

The Route A engineering line retains those boundaries while adding a closed qualification and evidence path. A six-job, permanently non-admissible qualification first exercises a synthetic producer/replay pair, then a case-shaped native producer with one discarded warm-up and three fresh-key recorded lanes, exact package replay with zero key generation and zero encryption, a cross-domain combined guard, and a post-run resource record. Its artifacts have one-day retention, carry no publication authority, and can never enter the paper’s formal result set. An external controller enforces the frozen computational and end-to-end deadlines and can only return a non-serializable, short-lived dispatch capability after a fresh provider reread.

Formal execution was specified as a separate strictly serial campaign, conditional on qualification success. Every unit would have had an experiment-source checkout, producer, one-day non-evidence handoff, independent replay, guard, and formal artifact. The terminal validator admits exactly one acquisition artifact, six synthetic artifacts, four ordered-event artifacts, and six native artifacts before one aggregate and a compatible detached analysis snapshot may be created. A clean source snapshot, a data-only registration snapshot, and an analysis snapshot may have different commit identifiers only when a repository-owned closed Behavior Set and compatibility receipt prove that every behavior-bearing blob remains identical where required. The qualification did not succeed, so none of these formal units or downstream objects was created.

Historical exact-head CI and real-runner smokes are useful engineering observations, but they do not transfer across behavior changes. The final reviewed tree was merged as Experiment Source Snapshot S1 ee58627bb5752c6ac1ee2c5132c6574f9cb66552; exact-main CI and PRE-S1 passed. A descriptive registration was then installed as the only data change in Evidence-Freeze Snapshot S2 c7ff6820d9323f1850c1c5c57fd9070db88db120, whose CI and closed-Behavior-Set compatibility checks passed. The sole qualification subsequently selected Route C, so the formal campaign, terminal admission, aggregate, and analysis chain were deliberately not run. No statement in this section is empirical evidence for a maintenance strategy.

A later, separately preregistered follow-up repaired a pre-dispatch capability- lease defect without changing the estimand, resource thresholds, or scientific matrix. Its replacement experiment source S1 was f8d89d6f98f289dc2e0c3414f7b4ed59b5d30f52, and its direct-child, data-only S2 was e1e488f177dc8a469c6132a29537b041fbf1430b. Exact-head CI, PRE-S1, descriptive registration, source-anchor, and independent-review controls all passed at exact S2 with authority false. The sole subsequent qualification run was cancelled before checkout or seed admission when the controller could not install its watcher-admission binding. It produced zero artifacts and no scientific observation. The frozen one-shot rule therefore closed the follow-up as terminal NO-GO and barred a repaired-lineage rerun.

Generative-AI systems assisted literature discovery, code and test generation, adversarial review, and drafting during development. Their outputs are neither source authority nor experimental evidence. Human authors must independently verify cited primary sources, inspect the final code, reproduce the admitted artifacts, and accept responsibility for every claim and released result.

6. Evaluation Methodology

Sections 6.1–6.6 preserve the frozen counterfactual execution and reporting protocol. Because neither qualification path produced a GO, none of the acquisitions, formal cells, native cases, measurements, or reports specified below occurred. The protocol was frozen before any formal artifact could be inspected. It specified no trained selector, winner chosen on a tuning prefix, or held-out oracle. Every strategy would have owned an independent persistent state and consumed the same ordered event groups, publication boundaries, queries, initial logical matrix, and public parameters. The intended intervention was therefore the maintenance mechanism, not the workload history.

6.1 Fixed strategies

The three strategy identities were:

1. periodic-repack/windows=1, which would have reconstructed and republished a complete static CSSC layout at every Publication Window;
2. padding-reuse, which would have reused the lowest-ordinal tombstone, then natural padding, and otherwise rebuilds the affected fixed horizontal row partition; and
3. packed-coo-cloud-segmented-delta/segment-width=128, which would have kept a CSSC base and appended overflow into cloud-executable, row-owned 128-lane segments. It was not to fold online in the registered experiment.

These candidates were mechanisms specified for comparison, not actions of an online policy. A slower strategy, a crossing cost curve, or the absence of a global winner would have been a reportable outcome rather than a failed experiment.

6.2 Synthetic matrix

The synthetic suite was specified to use a mixed insert/delete/modify workload at two frozen scales: S would have had 256 rows, 8,193 columns, and 512 accepted updates; M would have had 1,024 rows, 8,193 columns, and 2,048 accepted updates. The formal seeds were 20260822, 20260823, and 20260824, defining six scale– seed shards. Every shard would have evaluated all three strategies at $\rho \in \{0.01, 0.1, 1, 10\}$.

For a reduced fraction $\rho = p/q$ and zero-based accepted-event ordinal a, the number of queries inserted after event group a is

$$Q_a = \lfloor \frac{(a+1)p}{q} \rfloor - \lfloor \frac{ap}{q} \rfloor,$$

so the first N complete groups contain exactly

$$\sum_{a=0}^{N-1} Q_a = \lfloor \frac{Np}{q} \rfloor$$

queries. The 0.01, 0.1, and 1 cells were specified for direct execution. The $\rho = 10$ cell would not have been timed; it was registered as an exact transformation of the same strategy and shard's $\rho = 1$ query-linear count and byte fields. Any event, window, state, or non-whitelisted-field mismatch would have rejected the entire shard.

6.3 Ordered events from one real source

The sole external object would have been the SNAP Stack Overflow answer-to-question stream sx-stackoverflow-a2q.txt.gz (Stanford Network Analysis Project, n.d.; Paranjape et al. 2017). The acquisition job was specified to record the final URL, response headers, exact compressed length, and SHA-256 digest. An independent guard would have downloaded the object again and required identical response-body bytes; an admitted artifact would have omitted the raw compressed object.

The first 1,000,000 eligible records were specified to freeze the row/column mapping. Source-ID hashing would have defined two deterministic partitions, each with 1,024 rows and 8,193 columns. After the mapping prefix, each partition would have consumed 4,096 accepted records under two semantics. T1 was defined as cumulative occurrence,

$$A_{uv}(t) = min\{7, N_{uv}(t)\},$$

whereas T2 would have retained the most recent $K = 1024$ accepted events and performed expiry before admission in one indivisible atomic group. The synthetic logical clock would have advanced by one second per 128 accepted records. Both partitions and both semantics were specified to run all three strategies at $\rho \in \{0.1, 1\}$, yielding four ordered-event shards. If admitted, those shards could only have supported conclusions about deterministic interactions within one source, not multi-source or historical wall-clock generalization.

6.4 Native OpenFHE cases

The native matrix was specified to contain three strategies at the S and M scales, hence six cases. Every case would have used seed 20260822, $\rho = 1$, the terminal accepted-event prefix (512 for S and 2,048 for M), and the query after the last complete event group. Version, component inventory, query vector, OutputPlan, typed execution plan, and canonical input bytes were frozen together.

Each case was specified to perform one discarded warm-up, three fresh-key producer evaluations, and three exact package replays:

$$6 \times (1 + 3 + 3) = 42$$

native evaluations in total. The three producer repetitions specified for recording were technical repetitions, not independent population samples. An admitted execution would have reported their raw values, median, and range, separately for production and replay. Replay was specified to deserialize the retained context, secret/public keys, evaluation-key frame, and input ciphertexts. Its lifecycle receipt would have had to show zero context generation, zero key generation, zero evaluation-key generation, and zero encryption, while its cloud-program operation inventory would have had to equal that of the corresponding producer package.

6.5 Measurements and accounting

For strategy k, scale or source unit s, and query/update ratio ρ, the result was specified as a typed vector rather than a synthetic score:

$$c_{k,s,p}=\left(T_{state},T_{assembly},N_{op},B_{meta},\overline{B}_{crypto},RSS_{max},scratch_{max}\right).$$

Synthetic and ordered-event cells were specified to measure state-transition, result-assembly, and independent-replay time, peak RSS, and controlled scratch; they would have exactly counted events, windows, queries, typed operations, and object multiplicities. Their cryptographic-object bytes were specified as conservative type-derived bounds,

$$\overline{B}_{crypto}=\sum_{j} m_j U_j,$$

where m_j is the exact multiplicity and U_j is the registration-time, pre-execution Stage-1 maximum serialized size frozen for type j. Canonical metadata bytes would have been measured separately. Native cases instead were specified to report observed serialized-object bytes, typed-operation inventories, process time, RSS, and scratch. Simulator counts were never to be converted into OpenFHE latency, and evidence-replay overhead was never to be presented as strategy execution cost.

For the registered $\rho=10$ projection, only query-linear fields change:

$$n_{q,p}^{(10)}=10\,n_{q,p}^{(1)},\ n_{u,p}^{(10)}=n_{u,p}^{(1)}.$$

Wall time, RSS, scratch, and native latency would have been unavailable for this projected cell. Protocol-byte transfer time at bandwidth b Mbps was specified only as the transparent conversion

$$T_{net}(B,b)=\frac{8B}{b\times10^6};$$

HTTP/TLS, artifact wrappers, filesystems, and private replay transport were to remain separate evidence-pipeline costs.

The paired strategy contrast for unit u is

$$\tau_{a,b}(u,\rho)=C_a(E_u,\rho)-C_b(E_u,\rho),$$

where C is one preregistered field or the full typed cost vector. With only two scales, two deterministic source partitions, and three technical native repetitions, an admitted execution would have reported all raw points and mechanism-level decompositions. It would not have fitted scaling exponents, attached population p values or confidence intervals, or claimed a global winner or Pareto frontier.

6.6 Evidence admission and stopping

Before formal execution, one six-job qualification was required to complete within the frozen 45-minute computational path and 55-minute total path, satisfy the necessary $6C_q\leq9000\,s$ planning screen, and survive a fresh external-controller reread. Here C_q is the registered per-case native-runtime estimate:

the sum of provider-API startedAt–completedAt durations for the qualification's native producer job, native replay/guard job, and entire combined-guard job. Qualification artifacts were permanently non-evidence and could not enter the formal result set.

The formal campaign was specified as serial and would have admitted exactly

$$1\,\text{acquisition} + 6\,\text{synthetic} + 4\,\text{ordered-event} + 6\,\text{OpenFHE} = 17$$

pre-aggregate artifacts. Before launching each unit, the controller would have reserved its full remaining worst-case budget. The preregistered 12-hour figure was an acceptance threshold, not a guarantee that provider-side cancellation cannot overshoot. Across the 17 enumerated units, at most one whole-unit replacement would have been allowed, and only for a provider failure rather than a data-dependent or scientific outcome. Terminal admission would have rejected missing, extra, duplicated, wrong-attempt, or wrong-kind objects before aggregation and compatible detached analysis.

6.7 Current-source deterministic conformance replication

After both performance lineages were terminal, a separate preregistration froze a functional question that neither estimates nor rescues their performance estimand. The immutable source was lightweight tag current-source-e4-conformance-20260831-v1, pointing directly to reviewed commit 844fb062d78f5095f14599c6c71a27cb6034f001. The single default dispatch used .github/workflows/strong-whole-query-witness.yml, CPython 3.12, a 120-minute job timeout, two build threads, and OpenFHE 1.5.1 at commit 1306d14f8c26bb6150d3e6ad54f28dfe1007689e.

The deterministic property corpus froze seed 20260822, eight input cases, and 35 contract records. The encrypted witness froze one 4096-by-8193 CSSC-base-plus-strong-delta fixture, segment width 128, active payload 127, padding offset 127, global column 8192, and sparse expected output [(0, 128), (4095, 5)]. PASS required exact run/ref/head/attempt identity, success for every provider step, one success-named 30-day artifact, exactly 19 regular files, a raw ZIP digest equal to the provider digest, strict internal checksums, literal RFC 6901 checks, and detached-source PROVENANCE rehashing. The rerun button was forbidden; a replacement could exist only for a provider-only null attempt proven to have executed no job step and produced no artifact. No replacement was needed or dispatched.

7. Evidence Outcome

7.1 Evidence status

Exact-head CI, PRE-S1, descriptive registration, and S1–S2 compatibility gates closed before execution. Qualification run 33261434612 was then dispatched once from exact S2 while importing behavior from exact S1. At the frozen 45-minute computational threshold, the independent simulator replay was still running and the combined guard had not started. The external controller requested cancellation of that exact run. The terminal workflow conclusion is cancelled; the only retained object was the one-day q1 pre-replay handoff, which is permanently non-admissible.

No q5 guarded bundle, q6 post-run record, live dispatch capability, acquisition artifact, formal shard, terminal admission record, aggregate, or compatible S3 analysis exists. The project therefore has zero reportable strategy-cost, ordered-event, or native OpenFHE results. CI logs, qualification handoffs, partially completed jobs, and local copies are not alternative result sources.

Figure 2 records the evidence boundary rather than a performance curve. Green and blue boxes are engineering or workflow observations at their exact source identities; none is promoted into the absent formal result set.

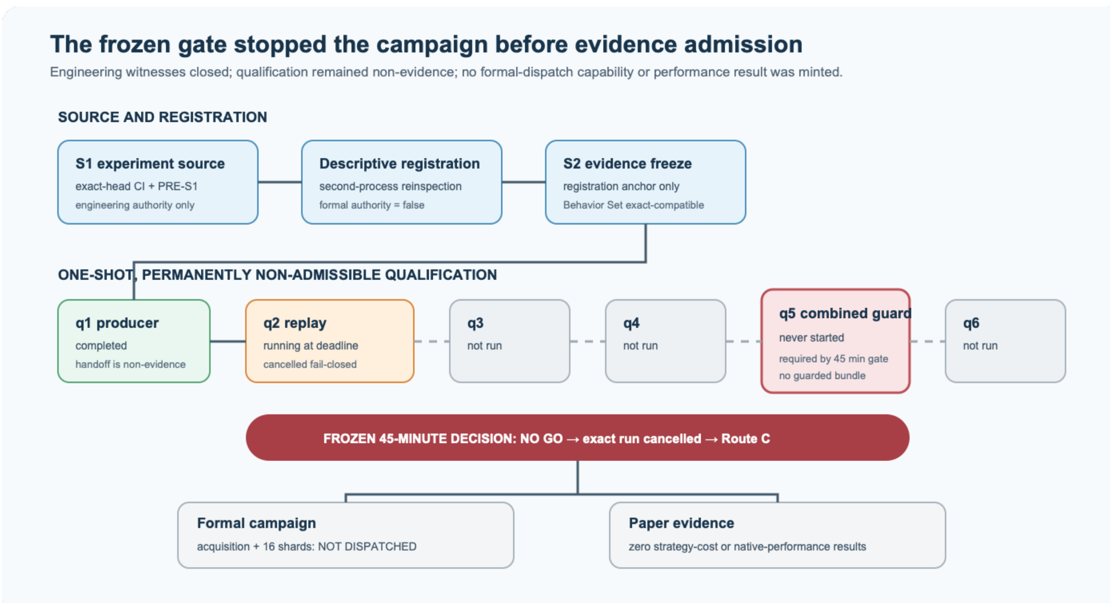

Figure 2. Primary exact-source freeze, one-shot qualification stop, and the resulting Route C evidence boundary; the later follow-up is recorded separately in Section 7.5.

Table 3. What each evidence layer proves and does not prove.

Layer	Exact disposition	Supports	Does not support
S1 CI and PRE-S1	Passed at exact S1 ee58627b…	Registered tests, source build, and ordinary/strong smoke execution at that identity	Strategy costs, complete-reference coverage, or formal artifact admission
Registration and S2	Descriptive archive reinspected; data-only anchor installed at c7ff6820…	Exact S1/S2 identity and closed-Behavior-Set compatibility	Qualification GO or a replayable dispatch authority

Layer	Exact disposition	Supports	Does not support
Qualification	One exact run; q1 completed, q2 was cancelled at the frozen stop, q5 never started	Preregistered Route C decision and fail-closed provenance	Any simulator/native performance estimator or partial formal result
Follow-up controls	Five fresh controls passed at follow-up S2 e1e488f…, all authority false	Exact follow-up source/anchor identity and prerequisite closure	Qualification GO, formal authority, or scientific execution
Follow-up qualification	One exact run was cancelled during hosted-runner setup before checkout or seed admission; zero artifacts	Terminal one-shot disposition and a control-plane failure chronology	Strategy behavior, performance, a 45/55-minute timeout, or a precise watcher/CAS root cause
Current-source E4 conformance	One exact tagged run and independently rehashed 19-file artifact passed	Version-bound eight-case/35-record contract outcome and one fixed OpenFHE whole-query vector	Candidate admission, complete-reference status, security, deployment, performance, speedup, or general correctness
Formal campaign	Not dispatched	The absence of unauthorized execution	Strategy ranking, speedup, ordered-event findings, or native resource claims

Table 4. Compact standalone provenance for the Route C stopping outcome.

Object or gate	Exact identity	Terminal disposition
Experiment source	S1 ee58627bb5752c6ac1ee2c5132c6574f9cb66552	CI run 33258436732 passed: 2,403 tests and 2 expected runner-dependent skips
Real-runner preparation	PRE-S1 run 33259569284	Passed: 583 tests and pinned OpenFHE 1.5.1 ordinary/strong smokes
Evidence freeze	Registration run 33259894587; S2 c7ff6820d9323f1850c1c5c57fd9070db88db120	Descriptive authority false; S2 CI run 33260167517 passed
One-shot qualification	Run 33261434612	cancelled; q1 completed, q2 stopped at the frozen gate, q3–q6 did not run, and q5 never started
Sole provider object	Artifact ID 9717884587, digest sha256:51cbfc2a…2c008	One-day q1 handoff; permanently non-evidence; no formal artifact exists

Table 5. Separately preregistered follow-up chronology.

Object or gate	Exact identity	Terminal disposition
Follow-up experiment source and evidence freeze	S1 f8d89d6f98f289dc2e0c3414f7b4ed59b5d30f52; S2 e1e488f177dc8a469c6132a29537b041fbf1430b; compatibility receipt c163974a…7e24	Direct-child data-only transition; all authority fields false
Five fresh controls	Runs 33347586537, 33347586567, 33347586568, 33347586620, and 33347586569	All completed successfully on exact S2; prerequisite evidence only
Sole follow-up qualification	Run 33348855548	completed/cancelled; only GitHub Set up job began; no checkout, registered seed, q1–q6 scientific stage, or artifact
Claim and evidence state	Qualification claim ref remains at exact S2; artifact API total_count=0; evidence root empty	Watch-binding candidate absent; no formal/campaign/terminal/analysis ref and no reportable result
Independent disposition	Same exact packet received two adversarial read-only reviews	Both returned TERMINAL-NO-GO; no second qualification or repaired-lineage continuation

The version-bound claim-ledger-draft.md supplement records the full provider digest, exact authority states, and the closed claim mapping. It is part of the circulation packet rather than an alternative source of empirical results.

7.2 Synthetic and ordered-event costs

The six synthetic and four ordered-event formal shards were not dispatched. Consequently this paper presents no strategy ranking, speedup, projected timing, or cost trade-off plot. The frozen schemas and analysis code remain part of the reproducibility package, but an empty or partial table is not treated as a measurement result.

7.3 Native OpenFHE execution

The six formal native cases were not dispatched. PRE-S1 ordinary and strong smokes demonstrate only that the pinned OpenFHE paths executed at the frozen source identity; they are not benchmark observations. No native median, range, speedup, memory result, or serialized-package comparison is claimed.

7.4 Preregistered Route C disposition

The qualification deadline failure is one of the preregistered falsifiers. It selects Route C without a threshold change, selective rerun, smaller matrix, or reuse of the partial handoff. GitHub's terminal metadata also reported two never-executed downstream cancelled jobs with completedAt one second earlier than startedAt; the controller rejected that terminal observation fail-closed. This provider-boundary anomaly did not create a false GO and did not cause the deadline failure. It is recorded as an implementation limitation for any future lineage, not as permission to modify S1 and repeat this one-shot attempt.

7.5 Separately preregistered follow-up disposition

The follow-up was designed after the primary stop to answer a narrower feasibility question without relaxing the original scientific matrix. Before its sole qualification, exact-S2 CI, PRE-S1, descriptive registration, source-anchor, and independent-review controls all completed successfully. One production controller call then consumed the nonserialized capability, created the qualification claim, and dispatched exact-S2 run 33348855548.

The controller failed closed while establishing the watcher-admission binding and cancelled the run. GitHub recorded only its internal Set up job step; no repository checkout, identity check, seed normalization, registered-seed computation, producer, replay, guard, or native stage ran. The other five jobs had zero steps, the artifact inventory was empty, and the claim ref remained at exact S2 rather than a watch-binding commit.

The frozen attempt definition attaches to the sole authorized provider dispatch, not to the first scientific step. The run’ s terminal non-success therefore exhausted the follow-up. Independent disposition reviews agreed that a direct rerun or a repaired watcher followed by replacement S1/S2 would be an impermissible outcome-informed second attempt. The exact failing substep remains unknown because the controller did not persist its nested exception chain; the unchanged ref alone cannot distinguish initial live-read construction, message-commit creation, or compare-and-swap failure. This uncertainty is a control-plane limitation, not evidence about a maintenance strategy.

7.6 Current-source deterministic conformance outcome

The independent controller audit accepted run 33386130654 at exact source 844fb062d78f5095f14599c6c71a27cb6034f001, attempt 1. Every provider-visible step succeeded. The only artifact was 9755741401, named strong-whole-query-witness-success-844fb062d78f5095f14599c6c71a27cb6034f001, with raw provider digest sha256:5978b7d9f75048939c9761243e224abb588ed82c2abc64e051523a7a598a1383. The downloaded ZIP independently reproduced that digest, contained exactly the 19 prespecified files, passed all strict checksums and JSON-pointer checks, and rehashed every provenance-listed source against a fresh detached checkout of the immutable tag.

At that exact source, the prespecified OpenFHE 1.5.1 whole-query witness for one fixed 4096-by-8193 CSSC-base-plus-strong-delta fixture decrypted to the prespecified sparse vector [(0, 128), (4095, 5)] and matched both independent plaintext oracles; all 35 records in the eight-case deterministic contract corpus passed. This is a single-source, single-fixture, version-bound functional observation. It is not candidate admission, a complete-reference claim, mixed-circuit safety, a security result, a deployment result, a performance measurement, or evidence that the stopped strategy comparison would have succeeded.

Table 6. Current-source E4 conformance provenance.

Object	Exact identity	Accepted observation
Immutable source	Tag current-source-e4-conformance-20260831-v1; commit 844fb062d78f5095f14599c6c71a27cb6034f001	Direct and peeled tag refs matched before dispatch and after completion
Provider run	33386130654; job 99469043424; attempt 1	completed/success; all 15 visible steps succeeded
Provider artifact	9755741401; 32,598 bytes; raw digest sha256:5978b7d9...a1383	One 30-day artifact; exact run/ref/head binding; 19/19 files and strict checksums passed
Property contract	Seed 20260822; eight input cases; 35 records	35 passed, zero failed; all authority flags remained false
OpenFHE witness	1.5.1 at 1306d14f...; fixed 4096-by-8193 fixture	Valid decryptions; [(0, 128), (4095, 5)]; typed and direct oracle matches true

8. Limitations

The threat model is narrow and provides no formal security theorem. Client B learns layout and reconstruction metadata; the Cloud learns shapes, counts, schedules, timing, opaque identifiers, and digests. The current witness covers one frozen fixture and one segment width. The real-stream matrix uses one source, two deterministic partitions, and two registered semantics; it does not support cross-domain generalization. The two synthetic scales do not identify an asymptotic exponent, and three fresh-key native repetitions do not identify a population distribution. The $\rho=10$ row is a registered count/byte projection, not a timing measurement. Unit microbenchmarks do not imply end-to-end latency or noise safety, and type-derived byte bounds are not observed ciphertext sizes. The present boundary is operational rather than a comparative strategy result: the formal matrix was never authorized. Post-outcome retuning, threshold changes, and selective reuse of qualification fragments are prohibited. The later follow-up also ended before repository or seed execution. It exposes a watcher-admission observability defect because the nested provider exception was not retained; it does not identify the failing substep, demonstrate a scientific timeout, or justify another attempt. Future independent studies should exercise and log that control-plane seam before consuming a one-shot capability. The separate current-source conformance result covers only one reviewed source, one fixed fixture, one segment width, and a deterministic eight-case corpus. It does not generalize across matrices, parameters, implementations, workloads, or security adversaries, and its provider timestamps are not performance data.

9. Conclusion

Mutable encrypted sparse computation requires more than an updatable container: matrix state, query reorganization, encrypted execution, and reconstruction must commit to the same version and be evaluated under complete causal costs. This work supplies that explicit contract around CSSC, three fixed maintenance mechanisms, and a fail-closed paired evaluation. The primary bounded attempt shows the consequence of treating feasibility gates as real falsifiers: when the qualification did not reach its combined guard by the frozen deadline, the system produced no authority and the formal campaign did not run. A separately preregistered follow-up later passed its prerequisite controls but failed while arming the one-shot watcher before scientific execution; its provider run and zero-artifact record likewise terminated without formal authority. A distinct, preregistered current-source conformance replication did pass one fixed OpenFHE whole-query fixture and its eight-case deterministic contract at an exact source identity. The resulting Route C manuscript therefore makes protocol, evidence-boundary, and narrowly version-bound functional claims, not comparative performance claims.

Data and code availability

The Route C package will identify the primary and follow-up S1/S2 pairs; their closed Behavior Sets and compatibility receipts; exact-head CI, PRE-S1, descriptive-registration, source-anchor, and independent-review identities; both qualification dispositions; the frozen schemas, functional propositions, source-conformance record, current-source E4 run 33386130654, artifact 9755741401, raw digest sha256:5978b7d9f75048939c9761243e224abb588ed82c2abc64e051523a7a598a1383, and verification code. It will explicitly state that the primary one-day q1 handoff was non-evidence, the follow-up produced no artifact at all, and no acquisition, formal, terminal, aggregate, or analysis artifact was created. The E4 artifact is functional conformance evidence, not a performance artifact. No archival DOI is claimed by this working draft. This section must be replaced with the accepted repository release and archival DOI before submission.

Statements and declarations

Funding. [The human author or authors must supply the applicable funding statement before submission.]

Competing interests. [The human author or authors must confirm and supply the competing-interest declaration before submission.]

Author contributions. [Insert the verified CRediT roles and author names; an AI system must not be listed as an author.]

Use of generative AI. Generative-AI systems assisted literature discovery, implementation and test generation, adversarial review, and manuscript drafting. Human authors must independently verify every source, inspect and run the released code and experiments, approve the final wording, and accept full accountability for the work. The exact disclosure will be rechecked against the publisher policy at submission.

References

The citation database for this working manuscript is references.bib. The venue-formatted manuscript must render every citation above and preserve DOI/arXiv links; this draft must not be submitted with citation keys left unresolved.

Bell, Nathan, and Michael Garland. 2008. *Efficient Sparse Matrix-Vector Multiplication on CUDA*. NVR-2008-004. NVIDIA Corporation. https://mgarland.org/files/papers/nvr-2008-004.pdf.

Bonawitz, Keith, Vladimir Ivanov, Ben Kreuter, et al. 2017. “Practical Secure Aggregation for Privacy-Preserving Machine Learning.” *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*, 1175–91. https://doi.org/10.1145/3133956.3133982.

Cash, David, Joseph Jaeger, Stanislaw Jarecki, et al. 2014. “Dynamic Searchable Encryption in Very-Large Databases: Data Structures and Implementation.” *Network and Distributed System Security Symposium*. https://www.ndss-symposium.org/ndss2014/ndss-2014-programme/dynamic-searchable-encryption-very-large-databases-data-structures-and-implementation/.

Chen, Qinghuan, Long Chen, Jimin Chen, Tao Li, Qinghong Cao, and Xiaoyang Zhou. 2026. “Retrieval Integrity Verification Mechanism with Privacy Protection and Dynamic Updates for Blockchain Oracles.” *Electronics* 15 (12): 2517. https://doi.org/10.3390/electronics15122517.

Crooks, Natacha, Matthew Burke, Ethan Cecchetti, Sitar Harel, Rachit Agarwal, and Lorenzo Alvisi. 2018. “Obladi: Oblivious Serializable Transactions in the Cloud.” *13th USENIX Symposium on Operating Systems Design and Implementation*, 727–43. https://www.usenix.org/conference/osdi18/presentation/crooks.

D’Agata, Lara, Carlos Agulló-Domingo, Óscar Vera-López, et al. 2026. “GPU Acceleration of Sparse Fully Homomorphic Encrypted DNNs.” *Proceedings of the Sixth European Workshop on Machine Learning and Systems*, 155–62. https://doi.org/10.1145/3805621.3807642.

Damie, Marc, Florian Hahn, Andreas Peter, and Jan Ramon. 2025. *Secure Sparse Matrix Multiplications and Their Applications to Privacy-Preserving Machine Learning*. https://arxiv.org/abs/2510.14894.

Fan, Junfeng, and Frederik Vercauteren. 2012. *Somewhat Practical Fully Homomorphic Encryption*. Cryptology ePrint Archive, Report 2012/144. https://eprint.iacr.org/2012/144.

Ferguson, Aidan, Perry Gibson, Lara D' Agata, et al. 2025. " Exploiting Unstructured Sparsity in Fully Homomorphic Encrypted DNNs." *Proceedings of the 5th Workshop on Machine Learning and Systems*, 31– 38. https://doi.org/10.1145/3721146.3721948.

Gao, Yang, Gang Quan, Wujie Wen, Scott Piersall, Qian Lou, and Liqiang Wang. 2026. "Efficient Privacy-Preserving Sparse Matrix-Vector Multiplication Using Homomorphic Encryption." *Information Sciences* 739: 123180. https://doi.org/10.1016/j.ins.2026.123180.

Halevi, Shai, and Victor Shoup. 2014. "Algorithms in HElib." In *Advances in Cryptology – CRYPTO 2014, Part i*, edited by Juan A. Garay and Rosario Gennaro, vol. 8616. Lecture Notes in Computer Science. Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-662-44371-2_31.

He, Jiaxing, Kang Yang, Guofeng Tang, et al. 2024. "Rhombus: Fast Homomorphic Matrix-Vector Multiplication for Secure Two-Party Inference." *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security*, 2490–504. https://doi.org/10.1145/3658644.3690281.

King, James, Thomas Gilray, Robert M. Kirby, and Matthew Might. 2016. "Dynamic Sparse-Matrix Allocation on GPUs." *High Performance Computing: ISC High Performance 2016*, Lecture notes in computer science, vol. 9697: 61– 80. https://doi.org/10.1007/978-3-319-41321-1_4.

Kreutzer, Moritz, Georg Hager, Gerhard Wellein, Holger Fehske, and Alan R. Bishop. 2014. "A Unified Sparse Matrix Data Format for Efficient General Sparse Matrix-Vector Multiply on Modern Processors with Wide SIMD Units." *SIAM Journal on Scientific Computing* 36 (5): C401–23. https://doi.org/10.1137/130930352.

Liu, Dongli, Wei Wang, Peng Xu, Laurence T. Yang, Bo Luo, and Kaitai Liang. 2024. " d-DSE: Distinct Dynamic Searchable Encryption Resisting Volume Leakage in Encrypted Databases." *33rd USENIX Security Symposium (USENIX Security 24)*, 2563–80. https://www.usenix.org/conference/usenixsecurity24/presentation/liu-dongli.

Mutluergil, Kemal, Deniz Elbek, Kamer Kaya, and Erkay Savaş. 2026. *Diagonal Packing for Efficient Homomorphic Sparse Matrix-Vector Multiplication*. https://arxiv.org/abs/2604.04683.

O'Neil, Patrick, Edward Cheng, Dieter Gawlick, and Elizabeth O'Neil. 1996. "The Log-Structured Merge-Tree (LSM-Tree)." *Acta Informatica* 33 (4): 351–85. https://doi.org/10.1007/s002360050048.

OpenFHE Development Team. 2026. *OpenFHE Development Repository*, *Version 1.5.1 Source Snapshot*. Git commit 1306d14f8c26bb6150d3e6ad54f28dfe1007689e. https://github.com/openfheorg/openfhe-development/tree/1306d14f8c26bb6150d3e6ad54f28dfe1007689e.

Paranjape, Ashwin, Austin R. Benson, and Jure Leskovec. 2017. "Motifs in Temporal Networks." *Proceedings of the Tenth ACM International Conference on Web Search and Data Mining*, 601–10. https://doi.org/10.1145/3018661.3018731.

Parbat, Tanusree, and Ayantika Chatterjee. 2023. " Authorized Update in Multi-User Homomorphic Encrypted Cloud Database." *IEEE Transactions on Knowledge and Data Engineering* 35 (8): 7796– 808. https://doi.org/10.1109/TKDE.2022.3221148.

Stanford Network Analysis Project. n.d. *Stack Overflow Temporal Network*. Stanford Network Analysis Project. https://snap.stanford.edu/data/sx-stackoverflow.html.

Stylianou, Chris, and Michele Weiland. 2022. *Exploiting Dynamic Sparse Matrices for Performance Portable Linear Algebra Operations*. https://arxiv.org/abs/2209.06478.

Vo, Viet, Xingliang Yuan, Shi-Feng Sun, Joseph K. Liu, Surya Nepal, and Cong Wang. 2023. " ShieldDB: An Encrypted Document Database with Padding Countermeasures." *IEEE Transactions on Knowledge and Data Engineering* 35 (4): 4236–52. https://doi.org/10.1109/TKDE.2021.3126607.

Wei, Yuntao, Xueyan Wang, Song Bian, Yier Jin, and Weisheng Zhao. 2026. *SparseE: Unlocking Sparsities in Encrypted Sparse Matrix Multiplication via Hardware-Software Co-Design*. 63rd Design Automation Conference Research Manuscript Program. https://63dac.conference-program.com/presentation/?id=RESEARCH2265\&sess=sess108.

Xiong, Wujie, Hao Zhou, Yutong Ye, Ruoming Jin, and Lei Xu. 2026. "CipherSkip: Efficient Sparse Matrix Multiplication with FHE." *Proceedings of the 40th ACM International Conference on Supercomputing*, 1220– 31. https://doi.org/10.1145/3797905.3807876.

Yu, Jiping, Kun Chen, Xiaoyu Fan, Yunyi Chen, Xiaowei Zhu, and Wenguang Chen. 2025. "Lodia: Towards Optimal Sparse Matrix-Vector Multiplication for Batched Fully Homomorphic Encryption." *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security*, 3649–63. https://doi.org/10.1145/3719027.3765025.